

The Master Quadratic of Discrete Spacetime: Deriving α^{-1} from Lattice Gauge Theory

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We derive the master quadratic $x^2 - 16(G^*)^2x + 16(G^*)^3 = 0$ from lattice gauge self-consistency, where $G^* = \sqrt{2}\Gamma(1/4)^2/(2\pi) \approx 2.9587$ is the lemniscatic constant. The roots $x_+ = 137.036$ and $x_- = 3.024$ match α^{-1} (1.26 ppm) and $N_c = 3$ (0.8%). We present the **xyztg ψ** ontology: the fundamental unit is the spacetime event (x, y, z, t) , constrained by geometry g and possibility ψ . Events are ontic. Constraints are real. Time emerges as ordering between events. Waves, motion, and probability distributions exist *between* moments, not within them. Wave function “collapse” is Bayesian updating by systems with ≥ 2 logic gates. The quadratic is a structural property requiring no observer.

THE XYZTG ψ ONTOLOGY

We seek the minimal ontic structure for physical reality without privileging observers or conflating events with laws.

The Three Primitives

Event (xyzt): The fundamental ontic unit. An irreducible fact: something happened here-now. A single event contains no motion, pattern, or wave. If $t = 0$, no event has occurred. A single event is complete.

Geometric Constraint (g): Encodes metric and causal structure. Constrains distances, durations, causal adjacency. Gravity is not a force but a constraint on spacetime relations. Inferred from relations among events, not directly observed.

Modal Constraint (ψ): Not an event or aggregate. A law-like constraint governing which events can occur with what relative likelihood. Does not “happen” at a moment; constrains transitions between moments.

The Single Moment

A single moment is fully described by:

$$\mathbf{xyztg}\psi \quad (1)$$

- $xyzt$: that an event occurred here-now
- g : geometric context making “here” and “now” meaningful
- ψ : modal constraint that made this event possible

Everything else exists *between* moments, not within them.

Time as Ordering

Time is not a primitive object or container. Time emerges as the ordering relation between events:

$$E_1 \prec E_2 \quad (2)$$

From ordering alone arise: before/after, duration, change, history. No observer is required for ordering to exist.

You can have (x, y) without t . You cannot have t without (x, y) .

Space can exist as static fact. Time requires substrate. Spatial existence is primitive; time is emergent.

Waves Exist Between Moments

A wave is not an instantaneous thing. It is a structured constraint on possible events across time. Wave-like behavior emerges only across multiple events.

At any single (x, y, z, t) , a particle either exists or does not. “Superposition” describes the ψ -constraint over multiple potential events, not a property of a single moment.

Minimal Causal Structure

Two events are connectable iff:

$$\text{Connectable}(E_1, E_2) \iff g(E_1, E_2) \wedge \psi(E_1 \rightarrow E_2) > 0 \quad (3)$$

Geometric admissibility (g): spacetime relation is allowed.

Modal admissibility (ψ): transition has nonzero amplitude.

Causality means: an event restricts events that may follow. No meaning or observer required.

Category	Gates	Substrate	Capability
Dead	0	None	None
Brain Dead	1	Any	Detection
Measurer	≥ 2	Inorganic	Inference
Observer	≥ 2	Organic	Inference

COMPUTATIONAL ONTOLOGY

0 gates: Exists at (x, y, z, t) but runs no update. A rock, a corpse.

1 gate: Boolean detection. Runs $\text{Detect}() \rightarrow \{0, 1\}$.

≥ 2 **gates:** Memory, comparison, inference. Runs the full:

$$\text{Entity}_{xyz t}.\text{Update}(\text{Input}, \text{Priors}, t, \text{action}) \quad (4)$$

Physics proceeds identically regardless of gate count.

Subjectivity and Objectivity

We all exist in $\text{Global.now}(t, e)$, where t is tick and e is entity. Objectivity is:

$$\text{Objective}(t) = \bigcup_e \text{Global.now}(t, e) \quad (5)$$

Subjectivity is local processing at entity e . Not mysterious: spatially separated updates on local information at shared t .

Observers and Meaning

Observers are not causally special. Measurement is physical interaction producing stable record. Meaning, inference, narrative are emergent cognitive properties, not ontic primitives. Physics proceeds independently of interpretation.

THE LATTICE ACTION

On cubic lattice $\mathbf{L} \subset \mathbb{Z}^3$:

$$S[\mathbf{J}] = \sum_t \sum_v \left[\frac{1}{2} |\partial_t \mathbf{J}|^2 - \frac{1}{2} |\nabla \mathbf{J}|^2 \right] \quad (6)$$

No observer terms. No ψ -collapse terms. Pure constraint dynamics.

DEGREE OF FREEDOM COUNTING

Theorem: On the $2 \times 2 \times 2$ lattice:

$$N_{\text{DoF}} = 24 - 7 - 1 = 16 \quad (7)$$

The Gauss constraint $\nabla \cdot \mathbf{J} = 0$ eliminates longitudinal modes.

VACUUM POLARIZATION

The polarization correction:

$$\Pi(x) = \frac{16(G^*)^3}{x} \quad (8)$$

The lemniscatic constant:

$$G^* = \frac{\sqrt{2}\Gamma(1/4)^2}{2\pi} = 2.9586751... \quad (9)$$

selected uniquely by complex multiplication at $j = 1728$.

THE MASTER QUADRATIC

Self-consistency $x = 16(G^*)^2 - \Pi(x)$ yields:

$$\boxed{x^2 - 16(G^*)^2 x + 16(G^*)^3 = 0} \quad (10)$$

Roots:

$$x_+ = 137.0358... \quad (\alpha^{-1} \text{ to } 1.26 \text{ ppm}) \quad (11)$$

$$x_- = 3.0236... \quad (N_c = 3 \text{ to } 0.8\%) \quad (12)$$

TABLE I. Comparison with experiment

Quantity	Predicted	Experimental	Error
α^{-1}	137.0358	137.0360	1.26 ppm
N_c	3.024	3	0.8%

BARYON MASSES

The composition constant $K_{\text{comp}} = m_e/\pi$:

$$M_p = \left(\frac{13}{\alpha} + 55 \right) m_e - \frac{m_e}{\pi} = 938.272 \text{ MeV} \quad (13)$$

$$M_n = M_p^{\text{geo}} + (\phi^2 - 12\alpha)m_e - \frac{m_e}{\pi} = 939.565 \text{ MeV} \quad (14)$$

$$M_\Delta = \left(\frac{17}{\alpha} + 81 \right) m_e - \frac{m_e}{\pi} = 1231.7 \text{ MeV} \quad (15)$$

TABLE II. Baryon mass predictions

Particle	Predicted	Experimental	Error
M_p (MeV)	938.2724	938.2720	0.4 keV
M_n (MeV)	939.5654	939.5654	< 1 eV
M_Δ (MeV)	1231.7	1232	0.03%

EMERGENCE HIERARCHY

From $xyztg\psi$, higher phenomena arise:

- **Motion:** ordered chains of events
- **Trajectories:** constrained event sequences
- **Dynamics:** rules over admissible chains
- **Interference:** competition among alternative chains
- **Classical behavior:** dominance of near-geodesic chains
- **Statistics:** many events over time
- **Subjectivity:** self-modeling event-processes

THE MEASUREMENT PROBLEM DISSOLVED

There is no measurement problem:

1. Events at (x, y, z, t) are primitive and definite
2. ψ constrains transitions, not moments
3. “Collapse” is $\text{Entity}_{xyzt}.\text{Update}()$
4. Physics runs whether any entity runs $\text{Update}()$ or not

A rock cannot collapse ψ because a rock cannot construct models from ψ .

CONCLUSION

The master quadratic is derived, not postulated. It produces α^{-1} to 1.26 ppm.

Core Principle: Reality consists of spacetime events constrained by geometry and possibility. Events are ontic. Constraints are real. Meaning is emergent.

The $xyztg\psi$ framework:

- $xyzt$: event (primitive)
- g : geometric constraint
- ψ : modal constraint

Everything else: motion, waves, probability, meaning, exists *between* moments.

Each of us runs $\text{Self}_{xyzt}.\text{Update}()$ within $\text{Global.now}(t, e)$. Subjectivity is local. Objectivity is $\bigcup_e \text{Global.now}(t, e)$.

The quadratic exists whether or not any $\text{Update}()$ runs to discover it.

Events are ontic. Constraints are real. Meaning is emergent. The lattice does not wait.

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